

```
In [99]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
df =pd.read_csv("D:/Downloads/Customer Churn.csv")
df.head()
```

Out[99]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Multipl
0	7590-VHVEG	Female	0	Yes	No	1	No	No
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	No
4	9237-HQITU	Female	0	No	No	2	Yes	

5 rows × 21 columns



```
In [2]: df.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   customerID            7043 non-null   object
 1   gender                7043 non-null   object
 2   SeniorCitizen         7043 non-null   int64
 3   Partner               7043 non-null   object
 4   Dependents            7043 non-null   object
 5   tenure                7043 non-null   int64
 6   PhoneService          7043 non-null   object
 7   MultipleLines         7043 non-null   object
 8   InternetService       7043 non-null   object
 9   OnlineSecurity        7043 non-null   object
10  OnlineBackup          7043 non-null   object
11  DeviceProtection      7043 non-null   object
12  TechSupport           7043 non-null   object
13  StreamingTV           7043 non-null   object
14  StreamingMovies       7043 non-null   object
15  Contract              7043 non-null   object
16  PaperlessBilling      7043 non-null   object
17  PaymentMethod         7043 non-null   object
18  MonthlyCharges        7043 non-null   float64
19  TotalCharges          7043 non-null   object
20  Churn                 7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

```

```

In [35]: df["TotalCharges"] = df["TotalCharges"].replace(" ", "0")
         df["TotalCharges"] = df["TotalCharges"].astype("float")

```

```

In [9]: df.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
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0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines           7043 non-null   object
8   InternetService        7043 non-null   object
9   OnlineSecurity          7043 non-null   object
10  OnlineBackup            7043 non-null   object
11  DeviceProtection        7043 non-null   object
12  TechSupport             7043 non-null   object
13  StreamingTV             7043 non-null   object
14  StreamingMovies         7043 non-null   object
15  Contract                7043 non-null   object
16  PaperlessBilling        7043 non-null   object
17  PaymentMethod           7043 non-null   object
18  MonthlyCharges          7043 non-null   float64
19  TotalCharges            7043 non-null   float64
20  Churn                   7043 non-null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1+ MB

```

```
In [10]: df.isnull().sum().sum()
```

```
Out[10]: np.int64(0)
```

```
In [11]: df.describe()
```

```
Out[11]:
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

```
In [12]: df.duplicated().sum()
```

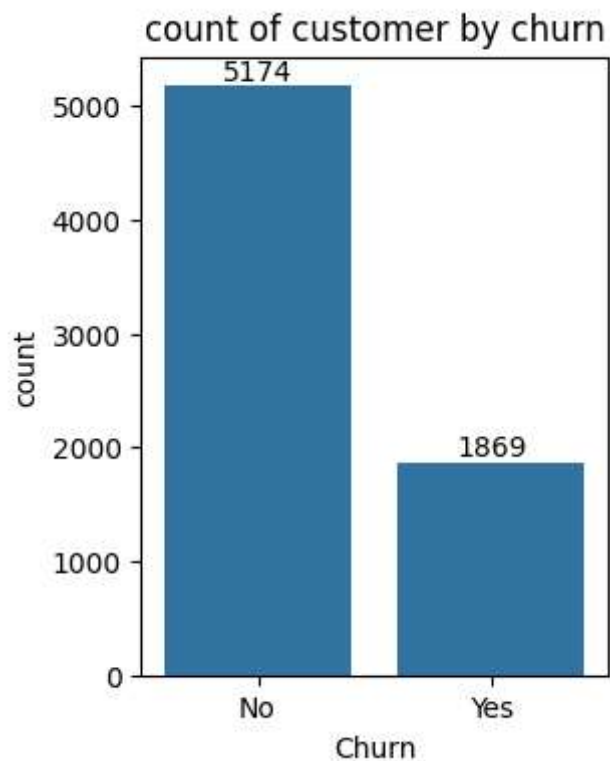
```
Out[12]: np.int64(0)
```

```
In [14]: df["customerID"].duplicated().sum()
```

```
Out[14]: np.int64(0)
```

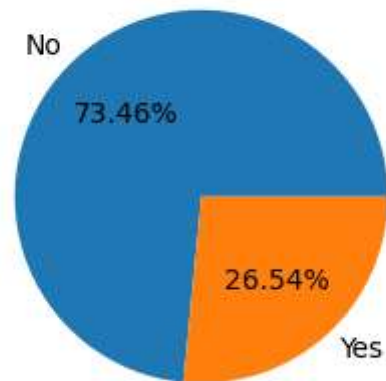
```
In [30]: def conv(value):  
    if value == 1:  
        return "yes"  
    else:  
        return "no"  
df["SeniorCitizen"] = df["SeniorCitizen"].apply(conv)
```

```
In [54]: plt.figure(figsize=(3,4))  
ax = sns.countplot(x="Churn",data =df)  
  
ax.bar_label(ax.containers[0])  
plt.title("count of customer by churn")  
plt.show()
```



```
In [53]: plt.figure(figsize=(3,4))  
gb =df.groupby("Churn").agg({"Churn":"count"})  
  
plt.pie(gb["Churn"],labels =gb.index, autopct ="%1.2f%%")  
plt.title("percentage of churned customers")  
plt.show()  
gb
```

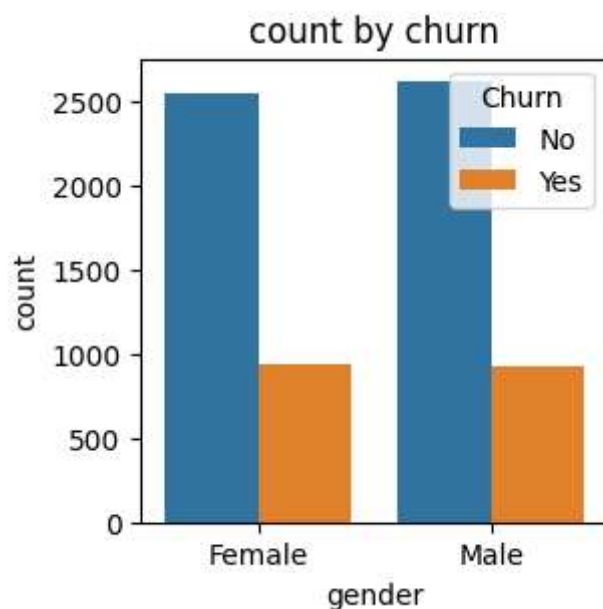
percentage of churned customers



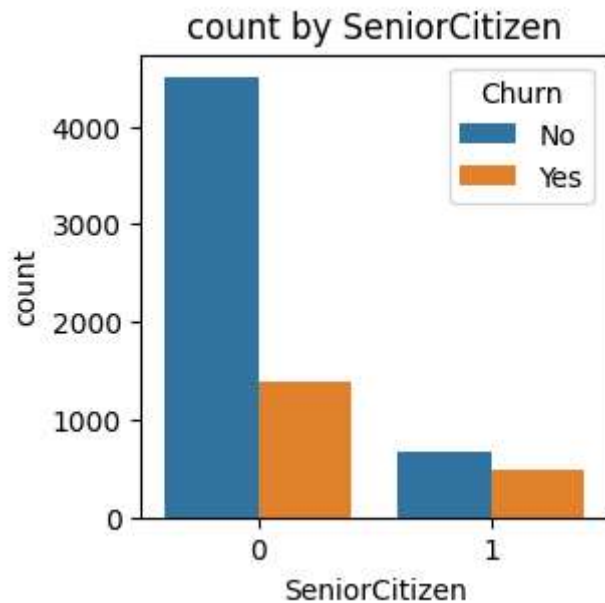
Out[53]:

Churn**Churn****No** 5174**Yes** 1869

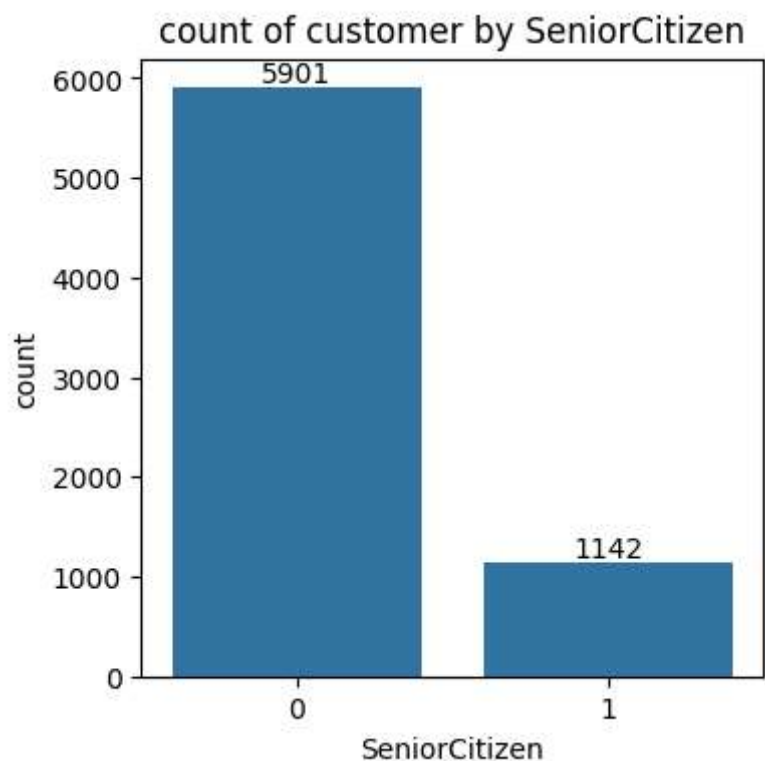
```
In [60]: plt.figure(figsize=(3,3))
sns.countplot(x="gender", data=df, hue="Churn")
plt.title("count by churn")
plt.show()
```



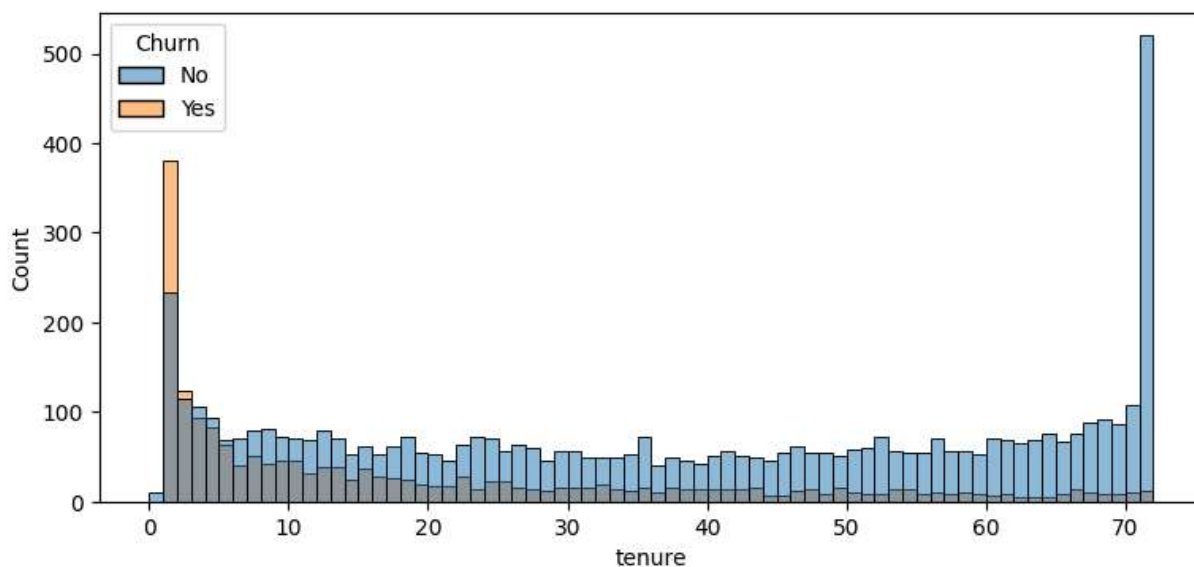
```
In [61]: plt.figure(figsize=(3,3))
sns.countplot(x="SeniorCitizen", data=df, hue="Churn")
plt.title("count by SeniorCitizen")
plt.show()
```



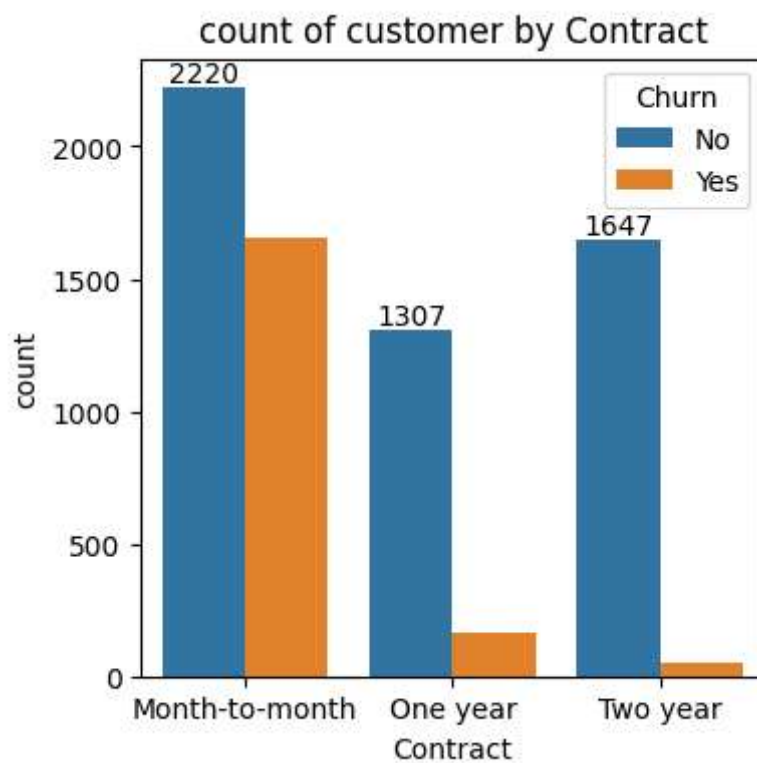
```
In [77]: plt.figure(figsize =(4,4))
ax =sns.countplot(x ="SeniorCitizen", data =df)
ax.bar_label(ax.containers[0])
plt.title("count of customer by SeniorCitizen")
plt.show()
```



```
In [74]: plt.figure(figsize =(9,4))
sns.histplot(x ="tenure", data =df,bins =72, hue ="Churn")
plt.show()
```



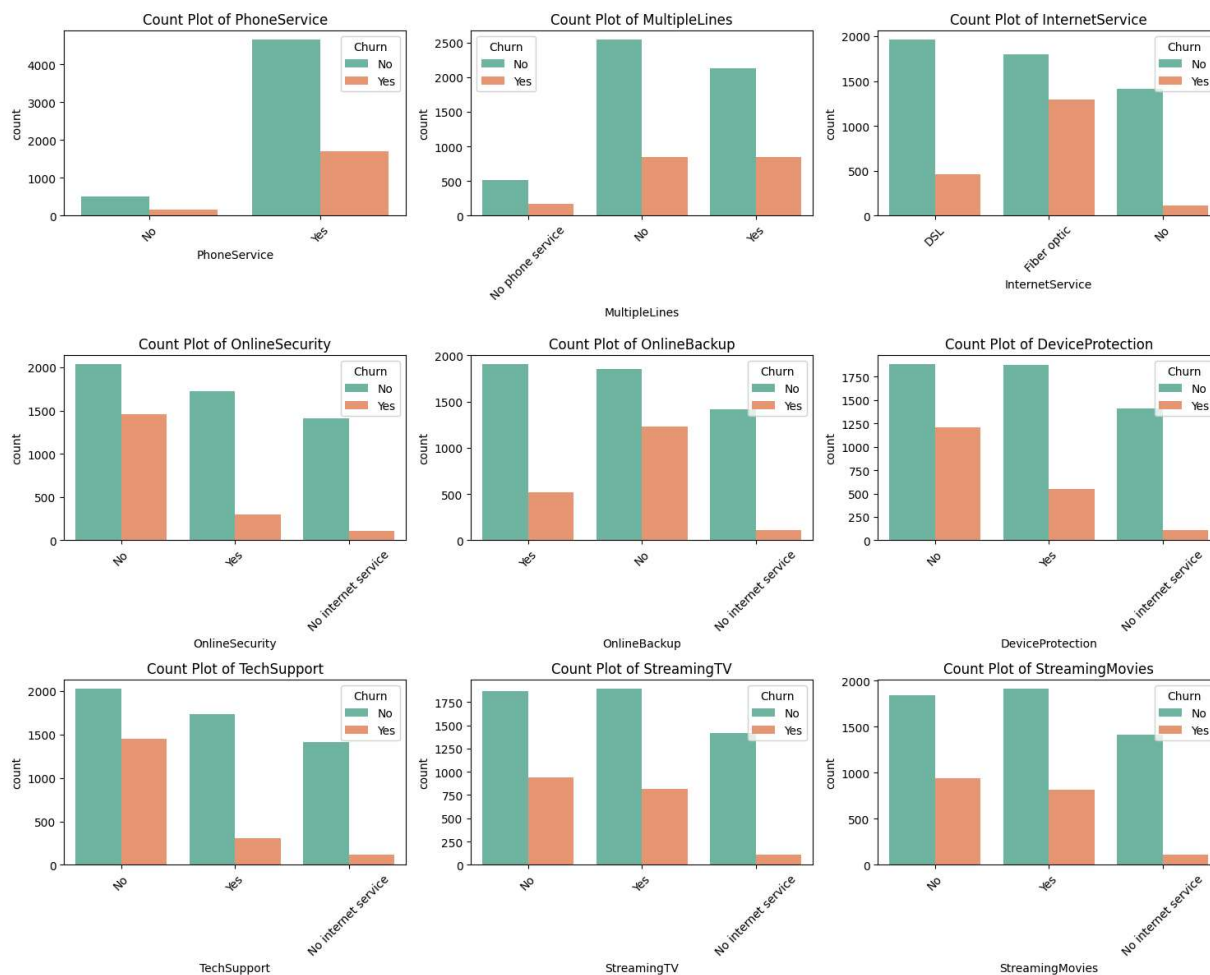
```
In [78]: plt.figure(figsize=(4,4))
ax = sns.countplot(x="Contract", data=df, hue="Churn")
ax.bar_label(ax.containers[0])
plt.title("count of customer by Contract")
plt.show()
```



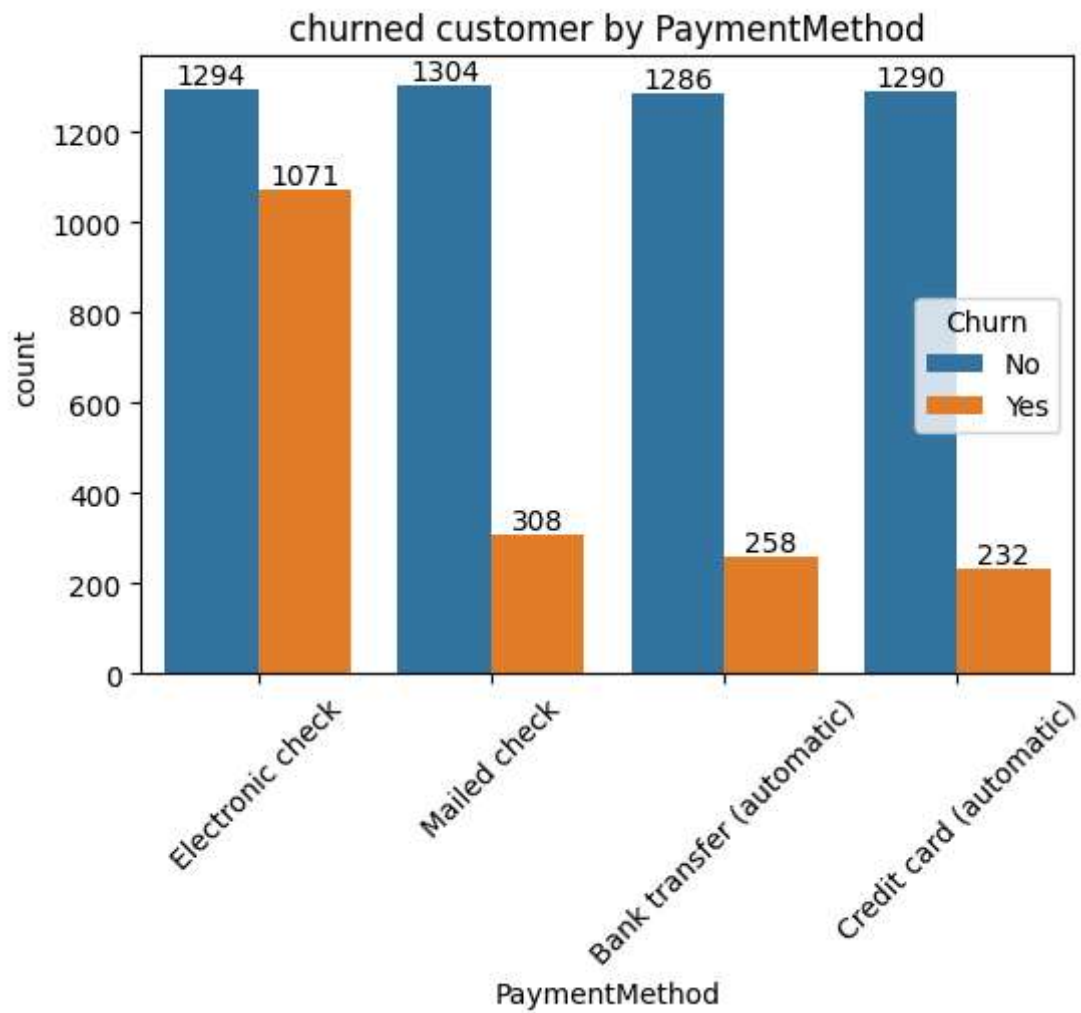
```
In [82]: df.columns.values
```

```
Out[82]: array(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',  
              'tenure', 'PhoneService', 'MultipleLines', 'InternetService',  
              'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',  
              'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',  
              'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',  
              'TotalCharges', 'Churn'], dtype=object)
```

```
In [93]: # Assuming your DataFrame is named 'df'  
cols = [  
    'PhoneService', 'MultipleLines', 'InternetService',  
    'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',  
    'TechSupport', 'StreamingTV', 'StreamingMovies'  
]  
  
# Set the number of rows and columns for subplots  
n_cols = 3  
n_rows = (len(cols) + n_cols - 1) // n_cols # Ceiling division  
  
plt.figure(figsize=(n_cols * 5, n_rows * 4)) # Adjust size as needed  
  
for i, col in enumerate(cols, start=1): # start=1 ensures i goes from 1 to 9  
    plt.subplot(n_rows, n_cols, i) # i must be an integer  
    sns.countplot(data=df, x=col, hue=df["Churn"], palette='Set2')  
    plt.title(f'Count Plot of {col}')  
    plt.xticks(rotation=45)  
  
plt.tight_layout()  
plt.show()
```

```
In [98]: plt.figure(figsize =(6,4))
ax =sns.countplot(x ="PaymentMethod", data =df,hue ="Churn")
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
plt.title("churned customer by PaymentMethod")
plt.xticks (rotation = 45)
plt.show()
```



In []: